

Conformity report of IPBES assessments with the data and knowledge management policy;

→ Invasive Alien Species

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Highlights

- The overall score of the Invasive Alien Species assessment was 93.2%
- Good compliance of most of the criteria, compared to the previous assessment, especially the delivery of figures and tables improved greatly.
- Instead of the improvement of compliance to the delivery protocol, this conformity assessment shows that a few criteria may be scored too harshly, specifically the criteria that all references need to have an identifier.

Rationale

To ensure that the assessments are delivered in agreement with the IPBES data and knowledge management policy, the delivery protocol for assessment drafts was created. An objective scorecard was previously designed to determine how well this protocol is followed. The first assessments scored according to this scorecard were the Values and Sustainable Use assessments, scoring 72.4% and 69.2% respectively. After the approval of the Invasive Alien Species at IPBES-10, the assessment report was scored following the same protocol, and received a final score of 93.2%. This report provides valuable insights on how well the IPBES data and knowledge management policy is currently implemented in IPBES products, and it highlights areas where the delivery protocol for assessment drafts could be improved.

Introduction

The first IPBES data management policy was published in 2020, after being approved by the MEP and Bureau meeting in January of that year. The IPBES delivery protocol of the assessment drafts was developed to assist the technical support units of the assessments with the implementation of the policy in their products. The protocol is a guideline for technical support units when preparing their milestone drafts to the Secretariat. The policy and delivery protocol are designed to be regularly updated, with the most recent revisions published in 2022 (data and knowledge management policy: <https://doi.org/10.5281/zenodo.3551078>; delivery protocol of the assessment drafts: <https://doi.org/10.5281/zenodo.6225417>).

To monitor the implementation of the policy within the Platform, the task force on knowledge and data requested its technical support unit to conduct a qualitative analysis of how well recently published IPBES assessments adhere to the policy, and to report the findings.

An objective scorecard was designed to ‘score’ each assessment on how well they complied to the delivery of the drafts protocol, and thereby to the data and knowledge management policy. The findings will give an insight into the performance of the assessments as well as show which (sub)sections of the policy and protocol are causing difficulties for the assessment TSUs. The technical support unit on data will use these findings to propose a revised version of the delivery of the assessment drafts protocol and provide adjusted support to the assessment TSUs which is scheduled for 2024 after the objectives for the Data and Knowledge Management Unit were approved at IPBES-10.

Results

The overall score of the Invasive Alien Species assessment was: 93.2%. The score of each of the subsections is broken down.

Citation: The assessment scored 100% for the citations. One important finding was that the authors list differed slightly from the list of experts on the IPBES website. As the authors list was compiled with all information directly collected from the authors themselves, the authors list was scored as being complete, but this does raise the important question on how these lists are compiled. To be able to use such lists for any analyses on the nationalities, countries of residence and countries of affiliation of the authors affiliated in IPBES, it is crucial that all country information is added by the authors directly. It is therefore recommended that the [guideline on experts lists](#) is being followed.

Bibliographic references: The assessment scored 85.7% for the bibliographic references, as not all entries in the bibliography have an identifier. The complete and clean Zotero library contains 5232 references. 3877 of the references have a doi (74.1% of all references), 474 references have an isbn number, 3306 references have an ISSN number and 4412 have an url (84.3% of the whole library). One reference has a web of science number. For 20 of the references the assessment tsu added a note that no identifier was available. 56 References did not have any identifier, and no note. In total, that means that 75 references did not have an identifier, which accounts for 1.4% of the entire library.

There were no retracted references in the (cleaned) library. All references cited in the assessment were included in the library, and all unused references were deleted from the library. For user-friendliness, tags for the different chapters were added, which makes it possible to see for which chapter(s) a reference was used.

Chapters and supplementary material: The assessment scored 100% on the delivery of the chapters and supplementary material. Each chapter and accompanying materials are uploaded to Zenodo, and connections/links between uploads are made on Zenodo. The docx files are linked to Zotero, and the bib files are added to the uploads on Zenodo.

Figures: The assessment scored 100% for the delivery of the figures. Because there were no country names used in the data, this criteria was excluded from the scoring. All the interoperable and final figures are uploaded to Zenodo. Official permission to use borrowed figures was requested and granted, and the captions of borrowed figures include the doi of the original source.

Tables: The assessment scored 66.7% for the delivery of tables. The tables and captions were uploaded to Zenodo. There are some tables in the assessment (table 4.26 on yield loss estimates due to *Spodoptera frugiperda* (fall armyworm) invasion and table 4.39 on the number of publications per country (top 12) on the impacts of invasive alien species using models and scenarios, and table 5.9 on examples of invasive alien species that were intentionally introduced for beneficial purposes, conflict resolution and potential management response) where the country names were written out instead of following the ISO 3166-1 alpha 3 code. The written out names almost always followed the country names as recommended by the United Nations, only in table 4.26 Tanzania is used instead of the correct name of United Republic of Tanzania.

Data and knowledge from indigenous people and local communities: All the requirements regarding data from Indigenous Peoples and Local Communities were fulfilled; they were informed, consent and approved the representation of their shared knowledge. The terms of agreement on reuse, representation and storage was documented and materials were uploaded to Zotero following these terms.

Disclaimers: The assessment fulfilled the requirements regarding the disclaimers; there is a general disclaimer for the full assessment, and a disclaimer for each individual chapter, which were checked and approved by the secretariat.

Discussion

By using a binary scoring metric, where a requirement is either fulfilled or not, the scoring is objective and simple, but sometimes this way of scoring can be too strict. For example, when scoring references for the presence of a unique identifier, there is no distinction between the lack of one identifier for the whole assessment or hundreds. It might be better to use percentages, for example, if one identifier is missing out of a total 8000 references, the score for this subsection would be 99.9%, which might be more informative than a 0. Especially for this assessment library that only contained 75 references without an identifier, a score of 0 does not seem fair. Also, as there were 3 tables with written out country names instead of the ISO 3166-1 alpha 3 code, its total score would have almost been 5% higher if the right code was used (98% instead of 93.2%). Allowing written out country names could be considered during the revision of the delivery protocol, to increase readability, but the official country names, as used by the UN, should be used. It would also be possible to have 2 columns, one with the official written out names, and one with the ISO 3166-1 alpha 3 codes.

Exporting and uploading bibliographies is a grey area. The protocol states that exporting should be done after the Plenary meeting. However, since there are typically six months between the Plenary meeting and the publication of the reports, one could expect that

when the reports are published the bibliographies should also be exported and uploaded. This could be made more explicit in the delivery of assessment drafts protocol, as 'after Plenary' leaves ambiguity on when this task should be completed. However, for this assessment this was not an issue, as the bibliographies were exported and uploaded in a timely manner.

It could also be suggested that the subsections should be delivered earlier. For example, it should not be possible that the complete authors list of an assessment is still incomplete after the publication of that assessment. The incongruence between the authors lists provided, the authors listed on the IPBES website and the authors in the citation of the chapter should be prevented.

Compared to the two assessments from the previous conformity report, the Values (72.4%) and Sustainable Use (69.2%) assessments, the Invasive Alien Species assessment scored notably higher. If the scoring of the identifiers and the use of country names in tables could be a little less strict, Invasive Alien Species would have scored 100%. Compared to the previous assessments, Invasive Alien Species did particularly better in the delivery of figures and tables. Working closely and cooperatively together with the tsu for invasive alien species proved to be very successful.

Based on this report, the task force on knowledge and data asked the technical support unit for data to revise the delivery protocol of the assessment drafts to increase its user friendliness.

Methods

The most recent version of the delivery protocol of the assessment drafts (<https://doi.org/10.5281/zenodo.6225417>) includes nine different sections:

- Citations (the citations of the full report, individual chapters, SPM and the authors list)
- Bibliographic references
- Supplementary material
- Delivery of schematic figures
- Delivery of data driven figures
- Delivery of borrowed/adapted figures
- Delivery of tables
- Delivery of data and knowledge from Indigenous people and local communities
- Disclaimers

To avoid a bias towards the figures, the sections delineating the delivery of figures were scored together, so that 7 sections remained. For each subsection, it was evaluated

whether the assessment complied with the requirements; and scored with a 1 if all elements of the subsection complied and scored with 0 if any element failed to comply. The total score of each section was transformed into a percentage, with a maximum score of 100%, after which the total score for the assessment was calculated as the average score of the different sections.

Scorecard for the Invasive Alien Species assessment

Description	Responsible Party	Binary	Count	Binary_sum	Score
			5	5	100.0%
Suggested citation entire assessment	Assessment TSU	1			
Suggested citation SPM	Assessment TSU	1			
Suggested citation individual chapters	Assessment TSU	1			
Complete list of authors	Assessment TSU	1			
Citations and list approved by Secretariat	Secretariat	1			
			7	6	85.7%
Library in Zotero	Secretariat	1			
Chapters in separate folders in Zotero	Secretariat	1			
All references cited entered in the right folder	Assessment TSU	1			
Unused references are deleted	Assessment TSU	1			
In-text citations complete and unbroken	Assessment TSU	1			
All entries have an identifier	Assessment TSU	0			
All references are in APA style	Assessment TSU	1			

			7	7	100.0%
Chapters in docx with connection to Zotero	Assessment TSU	1			
Chapters in pdf	Assessment TSU	1			
Exported bibliography (.bib)	Assessment TSU	1			
Authors' names in correct order	Assessment TSU	1			
Related identifier to SPM on Zenodo	Assessment TSU	1			
Related identifiers for DMR and data deposit packages	Assessment TSU	1			
Data management reports uploaded to Zenodo	Assessment TSU	1			
			18	18	100.0%
Original format uploaded to Zenodo	Assessment TSU	1			
Interoperable and open format uploaded to Zenodo	Assessment TSU	1			
Native files provided to graphic designer to update colours	Assessment TSU	1			

Country borders in line with UN guidelines	Assessment TSU	1			
Usage rights (CC0 or CC-BY) assigned	Assessment TSU	1			
Caption contains DOI and appropriate disclaimer if applicable	Assessment TSU	1			
Workflow/scripts used to create figure uploaded to Zenodo	Assessment TSU	1			
Native files provided to graphic designer to update colours	Assessment TSU	1			
Country borders in line with UN guidelines	Assessment TSU	1			
World maps in Robinson projection	Assessment TSU	1			
Country names in data follow the ISO 3166-1 alpha 3 code	Assessment TSU	-			
Usage rights (CC0 or CC-BY) assigned	Assessment TSU	1			
Caption contains DOI and appropriate disclaimer if applicable	Assessment TSU	1			

Permission to reuse figure is granted from original authors or publisher	Assessment TSU	1			
Figure is reviewed	Assessment TSU	1			
High resolution copy uploaded to Zenodo	Assessment TSU	1			
Proper citation of the figure added to Zotero library	Assessment TSU	1			
Usage rights (CC0 or CC-BY) assigned	Assessment TSU	1			
Caption contains DOI and appropriate disclaimer if applicable	Assessment TSU	1			
			3	2	66.7%
Original machine-readable data underlying the table along with caption uploaded to Zenodo	Assessment TSU	1			
Country names in data follow the ISO 3166-1 alpha 3 code	Assessment TSU	1			
Caption contains DOI and appropriate disclaimer if applicable	Assessment TSU	1			
			4	4	100.0%

IPLCs are informed and consent	Assessment TSU	1			
IPLCs approve the representation of their shared knowledge	Assessment TSU	1			
Terms of agreement on reuse, representation and storage is documented	Assessment TSU	1			
Materials are uploaded to Zotero following the agreed terms	Assessment TSU	1			
			3	3	100.0%
Suggested general disclaimer	Assessment TSU	1			
Suggested disclaimers for maps, figures and tables are added to captions when necessary	Assessment TSU	1			
Suggested disclaimers are checked and approved by Secretariat	Secretariat	1			
				Final Score	93.2%